



JAIPUR NATIONAL UNIVERSITY

Semester/Year End Examination Dec.2024

Name of School:	School of Engineering and Technology
Name of Programme / Course:	B.Tech. (Civil Engineering)/ B.Tech. (Computer Science)/ B.Tech. (Mechanical Engineering)/ B.Tech. CSE(Hons.) in Artificial Intelligence and Machine Learning/B.Tech. CSE(Hons.) in Cyber Security
Semester/Year:	3
Paper Code:	BTECECO304T22/BTECSCO301T22/BTEM ECO304T22
Title of Paper:	Mathematics III
Time: 3 Hours	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

This question paper consists of two sections, Section A and Section B. Section-A consists of seven questions and Section-B consists of six question. Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question.

Question Number	Question	Marks Allotted
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Section - A

- | | | |
|-------|---|---|
| Q. 1. | What is the expected number of heads of appearing when a fair coin is tossed 3 times. | 4 |
| Q. 2. | Differentiate between (M/M/1) : (∞ /FIFO) and (M/M/1) : (N/FIFO) | 4 |
| Q. 3. | Explain Karl Pearson coefficient of correlation. | 4 |
| Q. 4. | Write a Brief note on Curve fitting. | 4 |

- Q. 5. Write the procedure for Fitting of Second degree parabola 4
- Q. 6. Define Fourier transform and its inverse transform. 4
- Q. 7. Solve: $L\{\sin at\}$ 4

Section - B

- Q. 1. The distribution of number of letters received in an office for 100 days is given below. Use poisson distribution to find expected frequencies. 10

No. of Letters	0	2	3	4	5	6	7	8	9	10
Frequency	1	4	15	22	21	8	6	2	0	1

- Q. 2. Customer arrives at a box office window, being manned by a single individual according to a Poisson input process with a mean rate of 30 per hour. The time required to serve a customer has an exponential distribution with a mean of 90 seconds. Find the average: 10
- Waiting time of a Customer in the queue
 - Length of service
 - Time spent by customer
 - Queue Length

- Q. 3. In a beauty contest, three judges accorded following ranks to 10 participants: 10

Judge I	1	6	5	10	3	2	4	9	7	8
Judge II	3	5	8	4	7	10	2	1	6	9
Judge III	6	4	9	8	1	2	3	10	5	7

Interpret using Spearman's Rank Difference Method which pair of judges has a common taste in respect of beauty.

- Q. 4. Find the Inverse Laplace Transform of $\left[\frac{s+29}{(s+4)(s^2+9)} \right]$ 10
- Q. 5. Find fourier transform of $f(x) = \begin{cases} 1, & |x| \leq a \\ 0, & |x| > 0 \end{cases}$ 10
- Hence evaluate

4 a) $\int_0^{\infty} \frac{\sin s}{s} ds$

4 b) $\int_{-\infty}^{\infty} \frac{\sin s \cdot \cos sx}{s} ds$

4 Q. 6. Show that $\tan \theta$, the acute angle between 2 line of regression is given by

10

10
$$\tan \theta = \frac{(1 - r^2)}{r} \left(\frac{\sigma_x \cdot \sigma_y}{\sigma_x^2 + \sigma_y^2} \right)$$

And also interpret the case when $r = 0$ and $r = 1$

10